MAT3280 Sample solution to final exam

12/23/2023

Question 1. (10 points) Let $(X_n)_{n=1}^{\infty}$ be a sequence of independent random variables, and for $n \geq 1$, the distribution of X_n is given by

$$P(X_n = -n) = \frac{1}{2n}, \qquad P(X_n = 0) = 1 - \frac{1}{n}, \qquad P(X_n = n) = \frac{1}{2n},$$

(a) Does $(X_n)_{n=1}^{\infty}$ converge in probability?

(b) Does $(X_n)_{n=1}^{\infty}$ converge in mean square?

(c) Does $(X_n)_{n=1}^{\infty}$ converge almost surely?

Solution. (a) The random variables X_n, for $n \geq 1$, converge in probability to the constant 0. It is because for any $\epsilon > 0$, the probability $P(|X_n - 0| \geq \epsilon)$ is equal to $1/n$ for all $n > \epsilon$. Therefore, for any ϵ, the probability $P(|X_n - 0| \geq \epsilon)$ converges to 0 as $n \to \infty$.

(b) Because convergence in mean square implies convergence in probability, if X_n's converge in mean square, it'd better converge to the same limit as in part (a), namely, the constant 0. (We also use the a.s. uniqueness of limit in mean square and the uniqueness of limit in probability here.) So, we need to test whether X_n's converge to the constant 0 in mean square.

However, it cannot converge to 0 in mean square, because, for each n,

$$E[(X_n - 0)^2] = E[X_n^2] = \frac{1}{2n} \cdot \big(-n\big)^2 + \big(1 - \frac{1}{n}\big) \cdot 0^2 + \frac{1}{2n} \cdot \big(n\big)^2 = n,$$

which fails to converge to 0 as n increases. Therefore the answer to part (b) is no.

(c) Because convergence almost everywhere implies convergence in probability, if it converges, it would converge to the constant 0. However, the sequence does not converge to 0 almost everywhere. We can test the condition

$$\forall \epsilon > 0, \; P(|X_n| > \epsilon \;\; i.o.) = 0.$$

For a fixed $\epsilon > 0$, The probability that $|X_n|$ is larger than ϵ is equal to $1/n$ for all $n > \epsilon$. Since the sum of $1/n$ over all positive integers is a divergent, and because X_n are independent of each other, by the second Borel-Cantelli lemma, we have

$$P(|X_n| > \epsilon \;\; i.o.) = 1.$$

This proves that the sequence $(X_n)_{n=1}^{\infty}$ does not converge to 0 as $n \to \infty$. The answer to part (c) is also no.

Question 2. (15 points) Let $\tilde{X}$ be a Bernoulli random variable with success probability p, and let $\tilde{Y}$ be a Bernoulli random variable with success probability q. The probability p and q are distinct real numbers between 0 and 1. Assume that $p < q$ in this question.

(a) Find the joint distribution of two random variables X and Y, so that the marginal distributions of X and Y are the same as the distribution of $\tilde{X}$ and $\tilde{Y}$, respectively, and the probability of $P(X = Y)$ is as large as possible.

(b) Using the joint distribution in part (a), find the (nonlinear) minimum mean-square error estimation of X given Y.

Solution. The probability mass functions of $\tilde{X}$ and $\tilde{Y}$ are given by

$$P(\tilde{X} = 1) = p = 1 - P(\tilde{X} = 0)$$
$$P(\tilde{Y} = 1) = q = 1 - P(\tilde{Y} = 0)$$

We want to construct the joint distribution of two random variable X and Y, whose marginal distribution are the same as those of $\tilde{X}$ and $\tilde{Y}$, respectively.

	$X = 0$	$X = 1$
$Y = 0$	a	b
$Y = 1$	c	d

The joint distribution is specified by four nonnegative real numbers a, b, c and d, satisfying

$$\begin{aligned} a + c &= 1 - p, \\ b + d &= p, \\ a + b &= 1 - q, \\ c + d &= q. \end{aligned}$$

The constraints say that the entries in the second column of the table must sum to p, and the entries in the second row must sum to q. Because $p < q$, the largest value of d is equal to p.

We also have the requirements that the entries in the first column sum to $1 - q$ and the entries in the first row sum to $1 - p$. Since $1 - q$ is less than $1 - p$, the largest value of a is $1 - q$.

We can realize the largest values of a and d by setting the values of a, b, c and d as in the following table:

	$X = 0$	$X = 1$
$Y = 0$	$1 - q$	0
$Y = 1$	$q - p$	p

The value of $q - p$ is positive by the assumption that $p < q$. This gives the maximal coupling of $\tilde{X}$ and $\tilde{Y}$.

We can verify that this is indeed a maximal coupling by observing that probability $P(X \neq Y)$ is equal to $q - p$, which is exactly equal to the total variation distance of $\tilde{X}$ and $\tilde{Y}$.

(b) We want to estimate X as a function of Y. Let $\hat{X}$ denote the desired estimator,

$$\hat{X}(y) = \begin{cases} \alpha & \text{if } y = 0 \\ \beta & \text{if } y = 1, \end{cases}$$

here α and β are constant to be determined.

We solve for the constants α and β by applying the orthogonal principle, by using the indicator functions $\mathbf{1}_{\{Y=0\}}$ and $\mathbf{1}_{\{Y=1\}}$ are both $\sigma(Y)$-measurable.

From

$$\begin{aligned} \langle X - \hat{X}, \mathbf{1}_{\{Y=0\}} \rangle &= 0 \\ \langle \hat{X}, \mathbf{1}_{\{Y=0\}} \rangle &= \langle X, \mathbf{1}_{\{Y=0\}} \rangle \\ \alpha \cdot P(Y=0) &= 0 \cdot (1-q) + 1 \cdot 0 \\ \alpha \cdot (1-q) &= 0, \end{aligned}$$

we obtain $\alpha = 0$.

From

$$\begin{aligned} \langle X - \hat{X}, \mathbf{1}_{\{Y=1\}} \rangle &= 0 \\ \langle \hat{X}, \mathbf{1}_{\{Y=1\}} \rangle &= \langle X, \mathbf{1}_{\{Y=1\}} \rangle \\ \beta \cdot P(Y=1) &= 0 \cdot (q-p) + 1 \cdot p \\ \beta q &= p \end{aligned}$$

we obtain $\beta = p/q$.

The MMSE estimator of Y given X is

$$\hat{X}(Y) = \begin{cases} 0 & \text{if } Y = 0 \\ p/q & \text{if } Y = 1. \end{cases}$$

We may summarize the answer in one line by writing

$$\hat{X}(Y) = pY/q$$

The same answer can be obtained by considering the conditional expectation of X given Y, using the joint distribution of X and Y given in part (a). If we write the answer in terms of conditional expectation, the answer will look like

$$E[X|Y = y] = py/q.$$

Question 3. (15 points) Consider a sample space Ω and a function f mapping from Ω to $\mathbb{R}$.

(a) Prove that if f is measurable with respect to the trivial σ-algebra $\{\emptyset, \Omega\}$, then f is a constant function. (We suppose that the range $\mathbb{R}$ is equipped with the Borel algebra.)

(b) Suppose $\mathscr{F}$ is a σ-field generated by n subsets $A_1, A_2, \ldots, A_n$ in Ω. If f is $\mathscr{F}$-measurable, what is the largest number of distinct values that f can take? (Find the largest cardinality of the set $\{f(\omega) : \omega \in \Omega\}$. We note that the sets A_i's may not be mutually disjoint.)

Solution. (a) For each real number x, we consider the inverse image $E_x \triangleq f^{-1}(\{x\})$. The sets E_x, with x ranging over all real numbers are mutually disjoint. This is by the very definition of a mathematical function.

However, for each x, the inverse image E_x must be either $\emptyset$ or Ω, because it is assumed that each of them is $(\{\emptyset, \Omega\})$-measurable. Therefore, there is one and only one x_0 such that E_{x_0} is non-empty and is equal to Ω. The function f is the constant function that maps all ω in Ω to x_0.

(b) The atomic event in $\mathscr{F}$ has the form

$$B_1 \cap B_2 \cap \cdots \cap B_n,$$

where B_i is either equal to A_i or A_i^c, for $i = 1, 2, \ldots, n$. The function f takes a constant value within each atomic event in the above form. Since there are 2^n such events, the function f can take at most 2^n distinct values.

When $n = 0$, we have $2^0 = 1$, which is in accordance with the answer in part (a).

Question 4. (20 points) Let $(\Omega, \mathscr{F}, P)$ denote a probability space, and X be an integrable random variable. Suppose Ω is partitioned into a countable number of $\mathscr{F}$-measurable sets E_i, for $i = 1, 2, 3, \ldots$. (The sets E_i's are mutually disjoint, and the union of them is equal to the entire sample space Ω.)

(a) Prove that

$$\int X \, dP = \sum_{i=1}^{\infty} \int_{E_i} X \, dP.$$

(b) Let $\mathscr{G}$ denote the σ-algebra generated by the sets E_i, for $i \geq 1$. What is $E[X|\mathscr{G}]$?

(c) As a special case of part (b), suppose X is Gaussian distributed with mean 0 and variance σ^2, i.e., the pdf of X is given by

$$f_X(x) \triangleq \frac{1}{\sqrt{2\pi}\sigma} e^{-x^2/(2\sigma^2)}.$$

Let Y be the random variable obtained from X by rounding X to the nearest integer. What is $E[X|Y]$? (Write your answer in terms of $f_X(x)$. Numerical values are not required for this question.)

Solution. (a) For each positive integer m, we consider the random variable Y_m defined as

$$Y_m \triangleq X \cdot \sum_{i=1}^{m} \mathbf{1}_{E_i}.$$

The right-hand side is the same as multiplying X with the indicator function of the union $E_1 \cup E_2 \cup \cdots \cup E_m$. Hence, the value of $Y_m(\omega)$ is the same as X if ω is in $E_1 \cup E_2 \cup \cdots \cup E_m$. When ω is in the complement of $E_1 \cup E_2 \cup \cdots \cup E_m$, the value of $Y_m(\omega)$ is equal to 0.

We have $|Y_m(\omega)| \leq |X|$ for all ω, and for all m. Since the Lebesgue integral of $|X|$ is assumed to be finite, we can apply DCT to show that

$$\begin{aligned}
\sum_{i=1}^{\infty} \int_{E_i} X\,dP &\overset{(a)}{=} \lim_{m\to\infty} \sum_{i=1}^{m} \int X\mathbf{1}_{E_i}\,dP \\
&\overset{(b)}{=} \lim_{m\to\infty} \int \sum_{i=1}^{m} X\mathbf{1}_{E_i}\,dP \\
&\overset{(c)}{=} \int \lim_{m\to\infty} \sum_{i=1}^{m} X\mathbf{1}_{E_i}\,dP \\
&\overset{(d)}{=} \int X\,dP
\end{aligned}$$

where (a) follows from the definition of infinite sum, (b) from the linearity of integration, (c) from the dominated convergence theorem, and (d) from the assumption that $(E_i)_{i\geq 0}$ is a partition of Ω.

(b) The conditional expectation $E[X|\mathscr{G}]$ is a discrete random variable. It takes the value $E[X|E_i]$ when ω is in E_i. We express it as a summation in terms of indicator function

$$E[X|\mathscr{G}](\omega) = \sum_{i=1}^{\infty} E[X|E_i]\mathbf{1}_{E_i}(\omega) = \sum_{i=1}^{\infty} \Big(\frac{1}{P(E_i)} \int_{E_i} X\,dP\Big)\mathbf{1}_{E_i}(\omega)$$

(c) Let $\mathscr{G}$ denote the σ-algebra generated by Y. The atomic events in $\mathscr{G}$ are intervals

$$E_i \triangleq [i-0.5, i+0.5)$$

with n ranging over all integers. The other events in $\mathscr{G}$ are obtained by taking the union of some intervals in the above form.

For each i, the conditional expectation of X given E_i is

$$E[X|E_i] = \frac{\int_{i-0.5}^{i+0.5} x f_X(x)\,dx}{\int_{i-0.5}^{i+0.5} f_X(x)\,dx}.$$

This is the local average of X when restricted to the interval $[i-0.5, i+0.5)$. The conditional expectation of X given Y is the function

$$E[X|Y=i] = \frac{\int_{i-0.5}^{i+0.5} x f_X(x)\,dx}{\int_{i-0.5}^{i+0.5} f_X(x)\,dx},$$

where i ranges over all positive and negative integers.

Question 5. (20 points) We toss a fair dice repeatedly. Assume that the outcome of each throw are independent of each other. We model it by a sequence of i.i.d. random variables $(X_i)_{i=1}^{\infty}$ such that X_i is uniformly distributed on $\{1,2,3,4,5,6\}$, for each i.

(a) Let n be an integer larger than or equal to 3. What is the expected number of three consecutive 6's in the first n throws of the die. The counting is done with overlap. For instance, five consecutive 6's is counted as three triples of 6.

More precisely, we define Y_n as the number of indices i from 1 to $n-2$ such that $X_i = X_{i+1} = X_{i+2} = 6$. Compute the expectation of Y_n.

(b) Prove that the random variable Y_n/n converges in probability to a constant as n approaches infinity. (Hint: First prove that $Y_n/(n-2)$ converges to a constant in probability, and then show that Y_n/n and $Y_n/(n-2)$ converge to the same limit in probability as $n \to \infty$.)

Solution. (a) For $i = 1, 2, \dots, n-2$, let A_i denote the event that X_i and X_{i+1} and X_{i+2} are all equal to 6. Then

$$Y_n = \sum_{i=1}^{n-2} \mathbf{1}_{A_i}.$$

For each i, the expectation of $\mathbf{1}_{A_i}$ is the same as the probability $P(A_i)$, and is equal to $\frac{1}{216}$. Therefore, the expected value of Y_n is

$$E[Y_n] = \sum_{i=1}^{n-2} E[\mathbf{1}_{A_i}] = \frac{n-2}{216}.$$

(b) For notation convenience, we consider random variable

$$Z_i \triangleq \mathbf{1}_{A_i} - E[\mathbf{1}_{A_i}],$$

which is the centered version of random variable $\mathbf{1}_{A_i}$. By construction, the random variable Z_i has zero mean.

If indices i and j have difference larger than or equal to 3, the random variables Z_i and Z_j are independent. For $(i,j) = (i, i+1)$ and $(i,j) = (i, i+2)$, the pair of random variables Z_i and Z_j are correlated. However, it is not necessary to compute the covariance $Cov(Z_i, Z_j)$ exactly in the correlated cases. We only need to use an upper bound obtained by applying Cauchy-Schwarz inequality

$$Cov(Z_i, Z_j) \leq \sqrt{Var(Z_i) Var(Z_j)}.$$

Let σ^2 denote the variance of Z_1. Because Z_i's are i.i.d., σ^2 is also the variance of Z_i for all i. We thus obtain

$$Cov(Z_i, Z_j) \begin{cases} = 0 & \text{if } |j-i| \geq 3 \\ \leq \sigma^2 & \text{if } |j-i| \leq 2. \end{cases}$$

For $n \geq 4$, the variance of Y_n can be upper bounded by

$$\begin{aligned} Var\Big(Z_1 + Z_2 + \cdots + Z_{n-2}\Big) &= E[(Z_1 + Z_2 + \cdots + Z_{n-2})^2] \\ &= \sum_{i=1}^{n-2} Var(Z_i) + 2\sum_{i=1}^{n-3} Cov(Z_i, Z_{i+1}) + 2\sum_{i=1}^{n-4} Cov(Z_i, Z_{i+2}) \\ &\leq [n - 2 + 2(n-3) + 2(n-4)]\sigma^2 \\ &= (5n - 16)\sigma^2. \end{aligned}$$

This increases in the order of $O(n)$.

Hence, by Chebyshev inequality, for each $\epsilon > 0$,

$$P\Big(\Big|\frac{Y_n}{n-2} - \frac{1}{216}\Big| > \epsilon\Big) \le \frac{(5n-16)\sigma^2}{(n-2)^2\epsilon^2},$$

which tends to 0 as $n \to \infty$.

Therefore, $Y_n/(n-2)$ converges to the constant $1/216$ as $n \to \infty$.

On the other hand, we can show that for each $\epsilon > 0$,

$$P\Big(\Big|\frac{Y_n}{n-2} - \frac{Y_n}{n}\Big| > \epsilon\Big) = P\Big(2\frac{Y_n}{n(n-2)} > \epsilon\Big) \le \frac{2E[Y_n]}{n(n-2)\epsilon} = \frac{2}{216n\epsilon}$$

by Markov inequality. We have

$$P\Big(\Big|\frac{Y_n}{n-2} - \frac{Y_n}{n}\Big| > \epsilon\Big) \to 0$$

as $n \to 0$.

We now show that Y_n/n approaches $1/216$ in probability as $n \to \infty$. By the usual triangle inequality for real numbers, for each sample point ω, we have

$$\Big|\frac{Y_n(\omega)}{n} - \frac{1}{216}\Big| \le \Big|\frac{Y_n(\omega)}{n} - \frac{Y_n(\omega)}{n-2}\Big| + \Big|\frac{Y_n(\omega)}{n-2} - \frac{1}{216}\Big|.$$

For any fixed $\epsilon > 0$, if we can guarantee that both

$$\Big|\frac{Y_n(\omega)}{n} - \frac{Y_n(\omega)}{n-2}\Big| \text{ and } \Big|\frac{Y_n(\omega)}{n-2} - \frac{1}{216}\Big|$$

are less than or equal to $\epsilon/2$, then we have

$$\Big|\frac{Y_n(\omega)}{n} - \frac{1}{216}\Big| \le \epsilon.$$

Hence, by considering the contra-positive, if

$$\Big|\frac{Y_n(\omega)}{n} - \frac{1}{36}\Big| > \epsilon$$

we must have

$$\Big|\frac{Y_n(\omega)}{n} - \frac{Y_n(\omega)}{n-2}\Big| > \frac{\epsilon}{2} \text{ or } \Big|\frac{Y_n(\omega)}{n-2} - \frac{1}{216}\Big| > \frac{\epsilon}{2},$$

or both. In other words, the event

$$\Big\{\omega \in \Omega : \Big|\frac{Y_n(\omega)}{n} - \frac{1}{216}\Big| > \epsilon\Big\}$$

is a subset of the union

$$\Big\{\omega \in \Omega : \Big|\frac{Y_n(\omega)}{n} - \frac{Y_n(\omega)}{n-2}\Big| > \frac{\epsilon}{2}\Big\} \cup \Big\{\omega \in \Omega : \Big|\frac{Y_n(\omega)}{n-2} - \frac{1}{216}\Big| > \frac{\epsilon}{2}\Big\}.$$

By taking probability of these events, we get

$$P\Big(\Big|\frac{Y_n(\omega)}{n}-\frac{1}{216}\Big|>\epsilon\Big)\le P\Big(\Big|\frac{Y_n(\omega)}{n}-\frac{Y_n(\omega)}{n-2}\Big|>\frac{\epsilon}{2}\Big)+P\Big(\Big|\frac{Y_n(\omega)}{n-2}-\frac{1}{216}\Big|>\frac{\epsilon}{2}\Big).$$

Since both terms on the right converges to 0 as $n\to\infty$, the left-hand side also converge to 0 as $n\to\infty$.

Question 6. (20 points) In this question we model a queue using martingale. Suppose time is discretized to $0,1,2,3,\ldots$. Initially, there are $x_0>0$ customers in a queue. At time t, when the queue is non-empty, the first person in the queue leaves, and Y_t new customers join the queue. When the queue is empty at time t, then nobody leaves the queue but Y_t new customers join the queue. We assume that the random variables Y_t's, for $t\ge 1$, are independent and distributed according to the probability mass function

$$P(Y_t=k)=\begin{cases}0.3 & \text{if } k=0\\ 0.5 & \text{if } k=1\\ 0.2 & \text{if } k=2.\end{cases}$$

We recursively define a sequence of random variables $X_0,X_1,X_2\ldots$ by setting X_0 to be the constant x_0, and for $t\ge 1$, let

$$X_t=X_{t-1}+Y_t-1.$$

Let T be the first time instance such that $X_t=0$. The objective of this question is to find the expected value of T. For $t\le T$, the value of X_t models the number of people in the queue. (When $t>T$, the value of X_t may be negative, which has no physical meaning, but we don't care. We only make sure that for $t\le T$, the random variable X_t is indeed equal to the number of people in the queue.)

For $t\ge 1$, let $\mathscr{F}_t$ denote the σ-algebra generated by $Y_1,Y_2,\ldots,Y_t$. Let $\mathscr{F}_0=\{\emptyset,\Omega\}$.

(a) Show that T is a stopping time relative to the filtration $(\mathscr{F}_t)_{t\ge 0}$.

(b) Let λ denote the mean of Y_1. For $t=0,1,2,3,\ldots$, let Z_t be the random variable $X_t+(1-\lambda)t$, Prove that $(Z_t)_{t\ge 0}$ is a martingale relative to the filtration $(\mathscr{F}_t)_{t\ge 0}$.

(c) Find the expected time until the queue is empty for the first time. (You can directly apply the optional stopping theorem without giving the justification of why it can be applied.)

Solution. (a) We first note that the values of $X_1,\ldots,X_n$ are determined by $Y_1,Y_2,\ldots,Y_n$, and vice versa. We have

$$\sigma(X_1,\ldots,X_n)=\sigma(Y_1,\ldots,Y_n),$$

i.e., the two σ-field $\sigma(X_1,\ldots,X_n)$ and $\sigma(Y_1,\ldots,Y_n)$ encode the same information.

By definition, the random variable T is equal to t if $X_i>0$ for all $i=1,2,\ldots,t-1$ but $X_t=0$. For any fixed positive integer n, the event

$$\{T>n\}$$

is the event that $X_0,X_1,\ldots X_n$ are all strictly positive. We can thus write

$$\{T>n\}=\bigcap_{t=1}^{n}\{X_t>0\}.$$

Because X_t is $\mathscr{F}_n$-measurable, for $t \leq n$, the above event is $\mathscr{F}_n$-measurable. The event $\{T \leq n\}$ is an event in $\sigma(Y_1, \ldots, Y_n)$, for each $n \geq 1$. Therefore, T is a stopping time with respect to the filtration $(\mathscr{F}_t)_{t \geq 0}$.

(b) We check that the three axioms of martingale.
(1) The random variable $X_t + (1-\lambda)t$ is integrable, because for each t, the value $(1-\lambda)t$ is a constant and

$$E[X_t + (1-\lambda)t] = E[X_t] + (1-\lambda)t.$$

The expectation of X_t is finite for each t, because it is equal to

$$X_t = x_0 + \sum_{i=1}^{t}(Y_t - 1).$$

For each t, the random variable $[X_t]$ has finite integral.
(2) Since $(1-\lambda)t$ is a deterministic value, the random variable $X_t + (1-\lambda)t$ is measurable with respect to the σ-field $\sigma(X_1, X_2, \ldots, X_t)$, which is equal to $\mathscr{F}_t$.
(3) We need to show that Z_t is a version of the conditional expectation $E[Z_{t+1}|\mathscr{F}_t]$.

$$\begin{aligned} E[Z_{t+1}|\mathscr{F}_t] &= E[X_{t+1} + (1-\lambda)(t+1)|\mathscr{F}_t] \\ &= E[X_t|\mathscr{F}_t] + E[Y_{t+1}|\mathscr{F}_t] - 1 + (1-\lambda)(t+1). \end{aligned}$$

The first term $E[X_t|\mathscr{F}_t]$ equals X_t, because X_t is measurable respect to $\mathscr{F}_t$. The second term $E[Y_{t+1}|\mathscr{F}_t]$ equals $E[Y_{t+1}]$, since Y_{t+1} is independent with $Y_1, \ldots, Y_t$. We thus get

$$E[Z_{t+1}|\mathscr{F}_t] = X_t + \lambda - 1 + (1-\lambda)(t+1) = X_t + (1-\lambda)t = Z_t.$$

(c) We can apply optional stopping theorem and solve for $E[T]$ from the equation $E[Z_T] = E[Z_0] = x_0$. When the martingale is stopped, we have $X_T = 0$, and hence

$$x_0 = E[Z_T] = E[X_T + (1-\lambda)T] = (1-\lambda)E[T].$$

This gives

$$E[T] = \frac{x_0}{1-\lambda} = 10x_0.$$